Probability & Statistics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

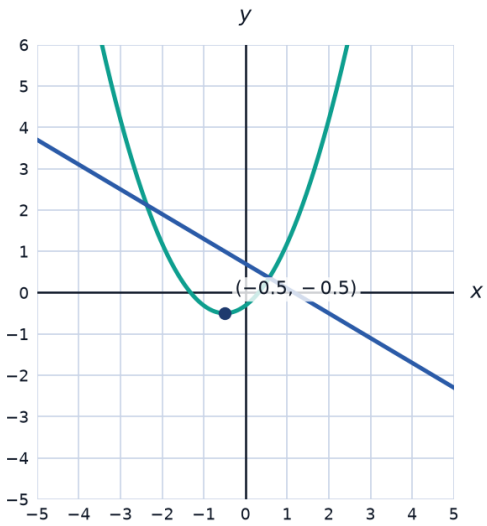
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the value of $16^{\frac{3}{4}}$?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which expression is equivalent to $(16x^8y^4)^{\frac{3}{4}}$ for all positive values of x and y?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What are the solutions to the quadratic equation $x^2 - 6x + 13 = 0$?

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What are the center and radius of the circle given by the equation $(x - 2)^2 + (y + 6)^2 = 36$?

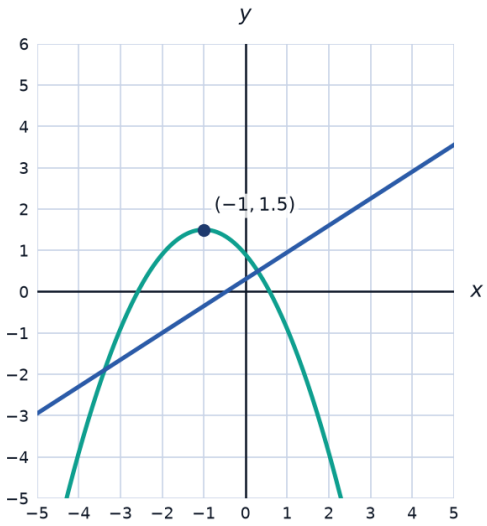
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A sequence is defined by the recursive formula $a_n = 2a_{n-1} + 1$, with the first term $a_1 = 3$. What is the value of the third term, a_3?

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?



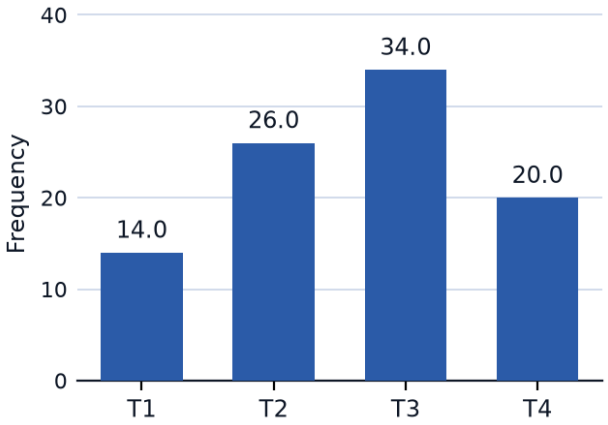
7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the center of the circle represented by the equation $(x - 3)^2 + (y + 5)^2 = 16$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

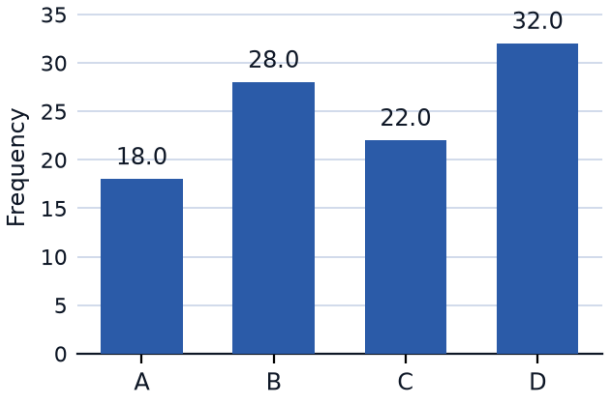
For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(5)$.



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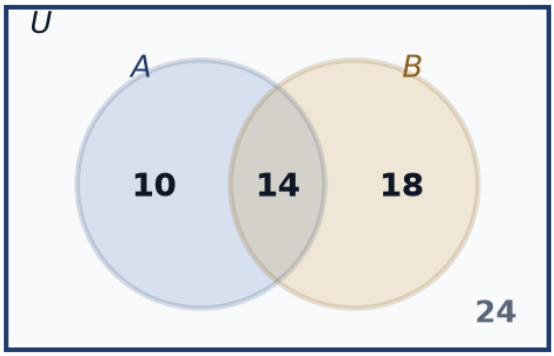
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(2)$.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for the normal distribution consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A committee of 4 members is to be selected at random from a club of 7 seniors and 5 juniors. (a) What is the total number of possible committees? (b) Find the probability that the committee contains exactly 2 seniors and 2 juniors.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a class of 200 students taking a test with mean 65 and SD 12 (normally distributed), how many students would you expect to score above 80?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the rational equation for x: $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$. Check for extraneous solutions.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 6.01 Radicals & Rational Exponents**

Solve the linear system using matrix inversion or Cramer's Rule: $3x - 2y = 8$ $5x + 4y = 6$

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 7.01 Sequences & Series**

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.



17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 8.01 Conic Sections**

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 9.01 Trigonometric Functions**

Solve $2x^2 - 6x + 5 = 0$ by completing the square, expressing all solutions in exact $a \pm bi$ form.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 1.01 Functions & Their Graphs**

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 10.01 Matrices & Systems**

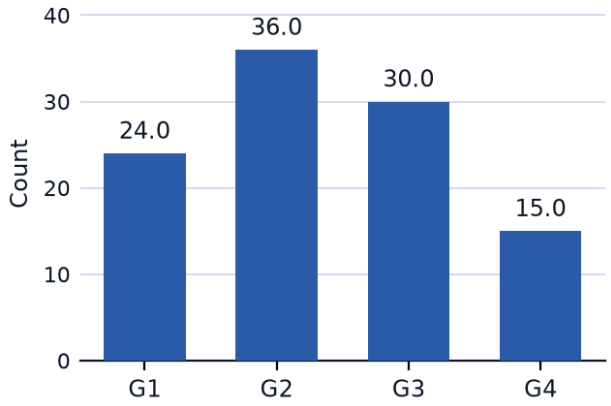
Solve the radical equation $\sqrt{2x+7} - x = 2$ for all real solutions and confirm extraneous roots.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(2)$.



a. 21

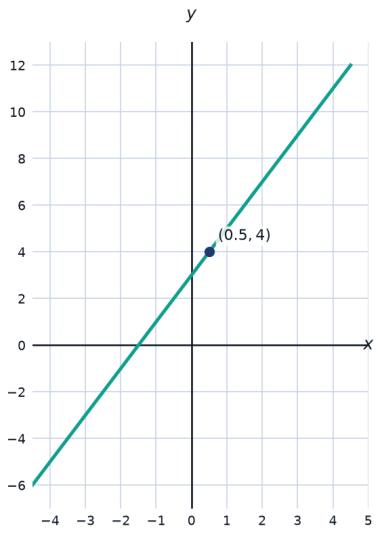
b. 27

c. 12

d. 25

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(2)$.



a. 7

b. 9

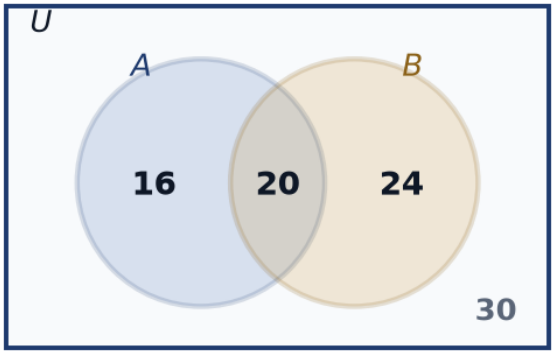
c. 4

d. 11

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for random sampling consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?



a. 20.5

b. 21.5

c. 22.5

d. 24.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for estimating proportions from samples consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for describing shapes of data distributions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for interpreting and comparing z-scores consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in interpreting and comparing z-scores, evaluate $f(3)$.

a. 21

b. 26

c. 15

d. 25

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for normal distribution probabilities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in normal distribution probabilities, evaluate $f(3)$.

a. 25

b. 31

c. 18

d. 29

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A fair six-sided die is rolled twice. Let X be the random variable representing the sum of the outcomes of the two rolls. The probability distribution of X is given in the table below, where p is an unknown probability. $Sum\ (x), 2, 3, 4, 5, 6, 7; P(X = x), 136, p, 336, 436, 536, 636;$ What is the value of p?

a. $\frac{2}{36}$

b. $\frac{5}{36}$

c. $\frac{4}{36}$

d. $\frac{3}{36}$

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The weights of apples harvested in an orchard are approximately normally distributed with mean μ and standard deviation σ. According to the empirical rule (68-95-99.7% rule), approximately 84% of the apples have a weight greater than 166 grams, and approximately 2.5% of the apples have a weight greater than 214 grams. What is the mean weight μ, in grams, of the apples?

a. 178 grams

b. 182 grams

c. 190 grams

d. 198 grams

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

In a championship playoff, two teams, Team A and Team B, play a best-of-five series, where the first team to win three games wins the series. The games are independent, cannot end in a tie, and in each individual game, the probability that Team A wins is p, where $0 < p < 1$ and $p \neq \frac{1}{2}$. Suppose the probability that the series is decided in exactly 3 games is equal to the probability that the series requires a decisive 5th game. What is the value of the product $p(1-p)$?

a. $\frac{\sqrt{33}-3}{12}$

b. $\frac{\sqrt{33}+3}{12}$

c. $\frac{\sqrt{33}-3}{6}$

d. $\frac{6-\sqrt{33}}{12}$

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A set of test scores is normally distributed with a mean of $\mu = 500$ and a standard deviation of $\sigma = 100$. According to the empirical rule (the $68--95--99.7\%$ rule), approximately what percentage of scores fall between 400 and 700?

a. 81.5%

b. 68.0%

c. 84.0%

d. 95.0%

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A factory produces two types of components, A and B, using two machines, M1 and M2. The time (in minutes) required to produce one unit of each component on each machine is given in the table below: | Component | Machine M1 | Machine M2 | |---|---|---| | A | 2 | 3 | | B | 4 | 1 | The factory has 480 minutes of M1 time and 240 minutes of M2 time available per day. Let x be the number of units of component A produced per day and y be the number of units of component B produced per day. Suppose the profit from selling one unit of component A is 5 and the profit from selling one unit of component B is 8. The factory wants to maximize its total daily profit. Formulate this problem as a linear programming problem, defining the objective function and the constraints.

a. Maximize $P = 5x + 8y$ subject to $2x + 4y \leq 480$, $3x + y \leq 240$, $x \geq 0$, $y \geq 0$.

b. Maximize $P = 8x + 5y$ subject to $2x + 3y \leq 480$, $4x + y \leq 240$, $x \geq 0$, $y \geq 0$.

c. Maximize $P = 5x + 8y$ subject to $4x + 2y \leq 480$, $y + 3x \leq 240$, $x \geq 0$, $y \geq 0$.

d. Maximize $P = 5x + 8y$ subject to $2x + 4y \leq 240$, $3x + y \leq 480$, $x \geq 0$, $y \geq 0$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A bag contains 5 red marbles and 3 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

a. $\frac{3}{8}$

b. $\frac{5}{8}$

c. $\frac{1}{2}$

d. $\frac{5}{3}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Prove the weak law of large numbers: $\bar{X}_n \to \mu$ as $n \to \infty$.

a. By Chebyshev's inequality: $P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0$. ✓

b. Not provable

c. Use the CLT only

d. Only for normal distributions

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The moment generating function uniquely determines the distribution. Why?

a. Only if $E(X)$ is finite

b. Not always true

c. Only for discrete distributions

d. Because the MGF encodes all moments and there is a one-to-one correspondence between MGFs and distributions (under regularity conditions)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Trigonometric Functions

State and prove the Bayes' theorem for continuous distributions.

a. Not valid for continuous distributions

b. Same as discrete case

c. $f(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)d\theta}$ — from the definition of conditional density

d. Use the CLT

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Functions & Their Graphs

Show that the binomial distribution approaches Poisson as $n \to \infty$, $p \to 0$, $np = \lambda$ fixed.

a. Requires $n > 100$

b. Not true

c. $\binom{n}{k}p^k(1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ — use $(1-\lambda/n)^n \to e^{-\lambda}$

d. Only for $k = 0$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Matrices & Systems

State the definition of a sufficient statistic.

a. $T(X)$ is sufficient for θ if the conditional distribution of X given $T(X)$ does not depend on θ

b. It must be unbiased

c. $T(X) = \theta$ always

d. Only the sample mean is sufficient

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

State and apply the Central Limit Theorem.

a. Only for discrete distributions

b. For large n, $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, regardless of the population distribution

c. Only for $n > 1000$

d. Only for normal populations

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Polynomial Functions

In a hypothesis test, the p-value is:

a. The probability that H_0 is true

b. The probability of observing a test statistic as extreme as observed, assuming H_0

c. The type II error rate

d. The significance level

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Rational Expressions

State the Fisher information: $I(\theta) =$

a. $E\left[\frac{\partial^2}{\partial\theta^2}\ln f\right]$

b. $E\left[\left(\frac{\partial}{\partial\theta}\ln f(X;\theta)\right)^2\right]$

c. $\frac{1}{\mathrm{Var}(\hat{\theta})}$

d. $\mathrm{Var}(\hat{\theta})$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Show the mgf of $\bar{X}$ for i.i.d. X_i with mgf $M(t)$ is $[M(t/n)]^n$.

a. $M_{\bar{X}}(t) = E[e^{t\bar{X}}] = E\left[\prod e^{tX_i/n}\right] = \prod M(t/n) = [M(t/n)]^n$

b. Only for normal X

c. Only for $n = 2$

d. Not true

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Radicals & Rational Exponents

For two random variables X, Y, prove $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$.

a. Only if independent

b. Only for discrete variables

c. $E[(X+Y-\mu_X-\mu_Y)^2] = E[(X-\mu_X)^2] + 2E[(X-\mu_X)(Y-\mu_Y)] + E[(Y-\mu_Y)^2]$. ✓

d. Not true

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